UNIT - IV

4 IoT Protocols

Syllabus

Protocol Standardization for IoT, M2M and WSN Protocols, RFID Protocol, Modbus Protocol, Zigbee Architecture. **IP based Protocols :** MQTT (Secure), 6LoWPAN, LoRa.

Contents

4.1 Protocol Standardization for IoT

- RFID and related identification techniques will be the key elements of the emerging IoT. A wide range of European Union (EU) funded RFID and IoT related projects have been set up in different application areas and are achieving considerable results. Communication, coordination and collaboration among these projects are essential for a competitive EU-funded research and for a secure and privacy-preserving deployment of RFID and IoT in Europe.
- To promote this communication, allow knowledge sharing and disseminate some of the relevant research results obtained in recent European research projects, the "Cluster of European RFID Projects" (CERP) was founded in January 2007. To reflect the current progress towards the IoT the name was changed to Cluster of European Research Projects on Internet of Things (CERP-IoT) in October 2008.
- Key objectives of the IoT- A consortium are as follows :
  1. Create the architectural foundations of an interoperable Internet of Things as a key dimension of the larger future Internet
  2. Architectural reference model together with an initial set of key building blocks.
  3. Federating heterogeneous IoT technologies into an interoperable IoT fabric.
- ITU-T study group are as follows :

Current Standard Activity		
ITU-T Study Group	**Study Group Name**	**Activities Related to IoT**
SG 2	Operational aspects of service provision and telecommunication management	Numbering, naming and addressing
SG 3	Tariff and accounting principles including related telecommunication economic and policy issues	--
SG 5	Environment and climate change	--
SG 9	Television and sound transmission and Integrated broadband cable networks	--

Current Standard Activity		
ITU-T Study Group	**Study Group Name**	**Activities Related to IoT**
SG 11	Signalling requirements, protocols and test specifications	Testing architecture for tag-based identification systems and functions
SG 12	Performance, QoS and QoE	--
SG 13	Future networks including mobile and NGN	NGN requirements and architecture for applications and services using tag-based ID
SG 15	Optical transport networks and access network infrastructures	--
SG 16	Multimedia coding, systems and applications	Requirements and architecture for multimedia information access triggered by tag-based ID
SG 17	Security	--
Pre-standard Activity		
Focus Groups	Smart Grid	Smart metering, M2M identify potential impacts on standards evelopment
	Cloud Computing	Cloud network requirements, e.g., for IoT
	Future Networks Car Communication	Describe future networks underlying the IoT Interaction of car hands free systems with the radio channel

- ITU can provide a forum to discuss these challenges, produce global standards and find global solutions to advance the Internet of Things and extend its benefits globally.
- Standards mean in general common methods, norms and regulations, based on which some work must be done, some product or service must be produced or some actions be conducted. Standards can be official and binding (de jure).

- Fig. 4.1.1 shows standardization process.

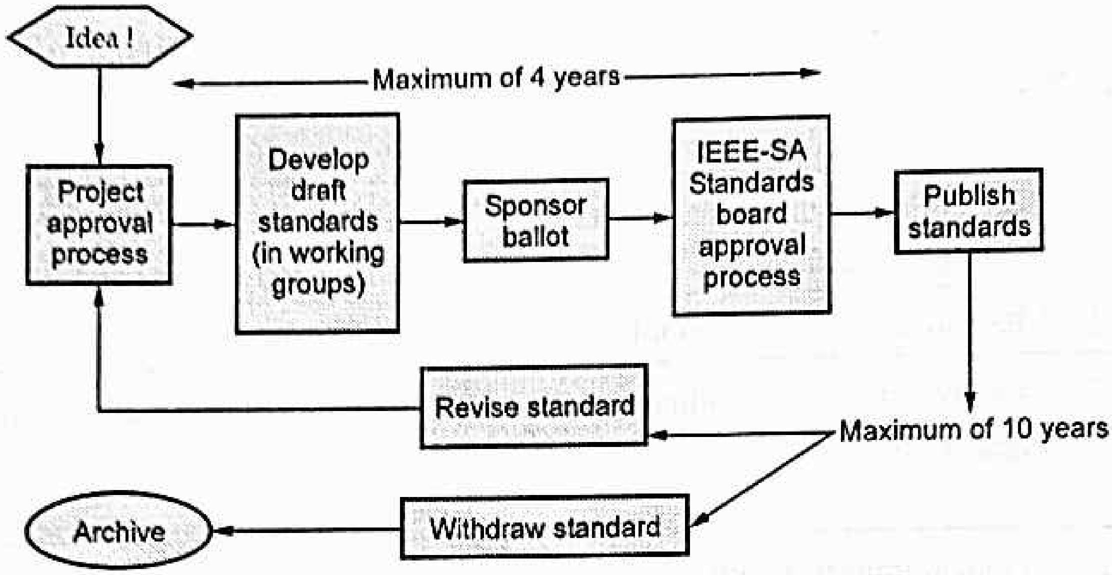

Fig. 4.1.1 Standardization process

- De facto standards can be formed by companies or groups of companies (interest groups, consortia, alliances, associations, etc.) which have come first into the market or application area and therefore the used methods/protocols etc. have become de facto standards. Standards play an important role in applying new technologies. With standards different actors in industry and in ecosystems can utilize similar and connective systems.
- Standards are published documents that establish specifications and procedures designed to maximize the reliability of the materials, products, methods, and/or services people use every day. Standards address a range of issues, including but not limited to various protocols to help maximize product functionality and compatibility, facilitate interoperability and support consumer safety and public health.
- **Sample list of ITU-T work related to IoT standardization**

SG13	**Y.2213**, NGN service requirements and capabilities for network aspects of applications and services using tag-based identification	Approved
SG13	**Y.2016**, Functional requirements and architecture of the NGN for applications and services using tag-based identification	Approved
SG16	**F.771**, Service description and requirements for multimedia information access triggered by tag-based identification	Approved
SG16	**H.621**, Architecture of a system for multimedia information access triggered by tag-based identification	Approved

SG17	**X.1171**, Threats and Requirements for Protection of Personally Identifiable information in Applications using tag-based identification	Approved
SG16	**H.IDscheme**, ID schemes for multimedia information access triggered by tag-based identification	On going work
SG16	**H.IRP**, ID resolution protocols for multimedia information access triggered by tag-based identification	On going work
SG11	**Q.nid-test-arch**, Testing architecture for tag-based identification systems and funtions	On going work

- Work package framework are listed below :

Work Package 1	Architectural Reference Model
Work Package 2	Orchestration and Integration into Future Internet Service Layer
Work Package 3	Protocol suite
Work Package 4	Resolution and Identification
Work Package 5	IoT Object Platforms
Work Package 6	Requirements, Validations and Stakeholder Interaction
Work Package 7	Use Cases
Work Package 8	Dissemination and Impact Generation
Work Package 9	Management and Coordination

- The emerging application space for smart objects requires scalable and interoperable communication methods which support future innovation as the application space grows. Internet protocol (IP) is suitable for this purpose. IP is highly scalable communication technology that supports wide range of applications and devices.
- An Internet protocol is open, lightweight, versatile and end-to-end communication method. It can runs on all types of devices like battery- operated embedded devices and tiny devices. therefore IP has all quality to make the Internet of Things a reality, connecting billions of communicating devices.
- The layered architecture of IP provides a high level of flexibility and innovation. IP provides standardized, lightweight, and platform-independent network access to smart objects and other embedded networked devices. The use of IP makes devices accessible from anywhere and from anything; general-purpose PC computers, cell phones, PDAs as well as database servers and other automated equipment such as a temperature sensor or a light bulb.

- IP runs over virtually any underlying communication technology, ranging from high-speed wired Ethernet links to low-power 802.15.4 radios and 802.11 (WiFi) equipment. For long-haul communication, IP data is readily transported through encrypted channels over the global Internet. Fig. 4.1.2 shows layered IP architecture.

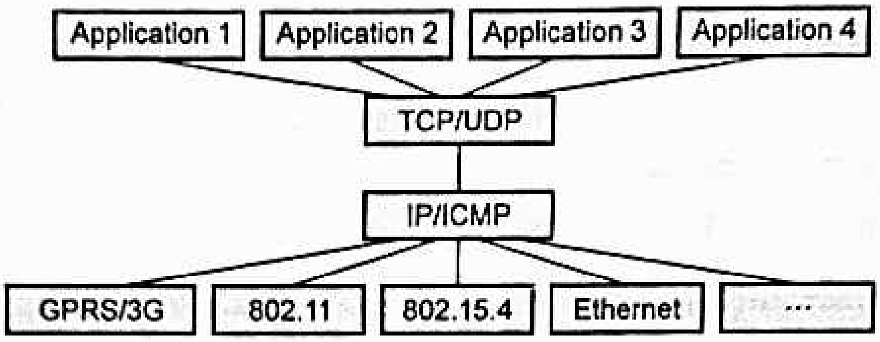

Fig. 4.1.2 : Layered IP architecture

- A smart object is defined by IPSO as an intelligent RFID tag, sensor, actuator and embedded device. The IPSO Alliance works closely with IETF, IEEE, ETSI and ISA relies on the standards developed by them.

4.1.1 M2M and WSN Protocols

- Currently available M2M applications are highly customized fashion. Auto industry and smart grid industry are try to expand their market vertically because of huge customer demands.
- A broad horizontal standard is a key requirement for the M2M industry to move from its current state of applications existing in isolated silos based on vertical market or underlying technology to a truly interconnected Internet of Things. Such a horizontal standard is expected to be the major growth in the future.
- Some of international organization like ITU, ETSI, Global Standards Collaboration (GSC) has established the M2M Standardization Task Force (MSTF, created during the GSC-15 meeting in Beijing, China, in September 2010) to coordinate the efforts of individual standards development organizations (SDOs), including China Communications Standards Association, Telecommunications Industry Association TR-50 Smart Device, etc.
- The high - level M2M architecture from MSTF does include fixed and other non-cellular wireless networks.
- The European Telecommunications Standards Institute (ETSI) produces globally-applicable standards for Information and Communications Technologies, including fixed, mobile, radio, converged, broadcast and Internet technologies.

- A new ETSI Technical Committee is developing standards for M2M Communications. This group aims to provide an end-to-end view of M2M standardization cooperating with ETSI's activities on Next Generation Networks and 3GPP standards initiative for mobile communication technologies.
- The ETSI M2M specifications are based on specifications from ETSI as well as other standardization bodies such as the IETF (Internet Engineering Task Force), 3GPP (3rd Generation Partnership Project), OMA (Open Mobile Alliance), and BBF (Broadband Forum).
- The Telecommunication sector of the International Telecommunication Union (ITU-T) has been active on IoT standardization since 2005 with the Joint Coordination Activity on Network Aspects of Identification Systems (JCA-NID), which was renamed to Joint Coordination Activity on IoT in 2011.
- The Open Geospatial Consortium (OGC 2013) is an international industry consortium of a few hundred companies, government agencies, and universities that develops publicly available standards that provide geographical information support to the Web, and wireless and location-based services.
- OGC Standards used to monitor, model and forecast flood, drought and environmental events. Government agencies can rapidly integrate data from different sensor networks. External (e.g. agricultural) sensor feeds can be incorporated easily to improve prediction and decision making.
- M2M standards activities include the following :
  1. Data transport protocol standards : M2MXML, JavaScript Object Notation (JSON)
  2. M2M device management
  3. M2M security and fraud detection
  4. Network API's M2M service capabilities
  5. Charging standards
  6. MULTI IMSI, M2M sevices that do not have MSISDN
  7. IP addressing issues for devices IPV6
  8. Remote diagnostics and monitoring, remote provisioning and discovery
  9. Remote management of devices behind a gateway or firewall
  10. Open REST- based API for M2M applications

- **China Mobile's WMMP standard**

Version	
1.	• It is based on Ericsson WMMP 0.98 platform • Support real time activity, user will manage and monitor M2M devices • Provides connection of devices to CMCC M2M Platform
2.	• Up-gradation in device protocol • It define management and business flow • It is based on Chongqing Radium M2M 2.0 platform
3.	• Include preamble definition, transfer process, device management, SIM card management, AT instruction format, etc. • Security measures are added • Add application API to platform. • Add device real time software upgrade function
Future	• Support wireless sensor network technology • Support non-standard and non-dedicated network and their protocols • Support distributedM2M network access • Realize automated terminal capacity distribution • Realize direct device-to-device communications

- Wireless Sensor Networks (WSN) are a special type of distributed measurement systems (DMS) whose use has been increasing in the last years particularly for monitoring, tracking and control.

- IEEE 1451 is a set of smart transducer interface standards developed by the IEEE Instrumentation and Measurement Society's Sensor Technology Technical Committee that describe a set of open, common, network- independent communication interfaces for connecting transducers (sensors or actuators) to microprocessors, instrumentation systems, and control/ field networks.

- IEEE 1415 provide a general transducer data, control, timing, configuration and calibration model. It provide a set of common interfaces for connecting sensors and actuators to instruments, and control and field networks.

- IEEE 1451.5 protocols are based on existing wireless protocols used for sensor networking. Specialized networks can handle only a limited number of sensor types or uses non-compact format. The 1451 is much superior at the sensor end.
- Most applications require individualized displays or graphical user interfaces, the 1451 is a fixed format and poorly suited at the user end. Network oriented applications prefer XML or similar formats which are convenient, but are too verbose at the sensor end. 1451 at the sensor end (Sensor Fusion level 0) combined with translators is the best solution.
- The US government may mandate a sensor data standard and the NIST-supported IEEE 1451 is the most recognized candidate. The sensor industry, especially the wireless network sector, must recognize the business advantages of a single sensor data standard.
- The IEEE 1451 family of standards includes the following :

IEEE 1451 standards	Functions
IEEE 1451.0-2007	Common Functions, Communication Protocols, and TEDS Formats
IEEE 1451.1 -1999	Network Capable Application Processor (NCAP) Information Model for smart transducers -- Published standard.
IEEE 1451.2 -1997	Transducer to Microprocessor Communication Protocols and Transducer Electronic Data Sheet (TEDS) Formats – Published standard
IEEE P1451.3	Digital Communication and Transducer Electronic Data Sheet (TEDS) Formats for Distributed Multi-drop Systems
IEEE P1451.4	Mixed-mode Communication Protocols and Transducer Electronic Data Sheet (TEDS) Formats
IEEE 1451.5-2007	Wireless Communication Protocols & TEDS Formats
IEEE 1451.7-2010	Transducers to Radio Frequency Identification (RFID) Systems Communication Protocols and TEDS Formats

- The main goal of the IEEE 1451 standard was to define a set of common communication interfaces to standardize the connectivity of transducers (sensors and actuators) to instruments, instrumentation systems, or control/field networks, which allow the access of transducers via local area networks or the Internet.
- IEEE 1451 standardizes two interfaces: a transducer device interface and a transducer network interface.

- IEEE 1451 also defines a set of metadata, called Transducer Electronic Data Sheets (TEDS). TEDS contain manufacture-related information about the transducers, such as manufacturer ID, serial number, measurement ranges, calibration data, location information, and more. TEDS allow the self-identification and self-description of transducers to the system or network.
- The sensor Web is a special type of Web-centric information infrastructure for collecting, modeling, storing, retrieving, sharing, manipulating, analyzing, and visualizing information about sensors and sensor observations of phenomena.
- Semantic Sensor Web (SSW) in which a sensor data is annotated with semantic metadata to increase interoperability as well as provide contextual information essential for situational knowledge.
- The OGC recently established Sensor Web Enablement (SWE) to address by developing a suite of specifications related to sensors, sensor data models, and sensor Web services that will enable sensors to be accessible and controllable via the Web.3.
- The European Union SENSEI project creates an open, business driven architecture that fundamentally addresses the scalability problems for a large number of globally distributed wireless sensor and actuator networks (WSAN) devices. It provides necessary network and information management services to enable reliable and accurate context information retrieval and interaction with the physical environment.
- **Tangible results of the SENSEI project are :**
  1. A highly scalable architectural framework with corresponding protocol solutions that enable easy plug and play integration of a large number of globally distributed WS&AN into a global system - providing support for network and information management, security, privacy, trust and accounting.
  2. An open service interface and corresponding semantic specifications to unify the access to context information and actuation services offered by the system for services and applications.
  3. Efficient WS&AN island solutions consisting of a set of cross-optimized and energy aware protocol stacks including an ultra-low power multi-mode transceiver architecture targeting 5nJ/bit.
  4. A pan-European test platform, enabling large scale experimental evaluation of the SENSEI results and execution of field trials - providing a tool for long term evaluation of WS&AN integration into the Future Internet.

- **ISO/IEC JTC 1/WG 7 Sensor Networks (WGSN)** is a standardization working group of the joint technical committee ISO/IEC JTC 1 of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), which develops and facilitates standards within the field of sensor networks. The international secretariat of ISO/IEC JTC 1/WG 7 is the Korean Agency for Technology and Standards (KATS), located in the Republic of Korea.
- The architecture is defined through the following set of documents :
  1. ISO/ IEC 29182 Part 1: General overview and requirements
  2. ISO/ IEC 29182 Part 2: Vocabulary/ terminology
  3. ISO/ IEC 29182 Part 3: Reference architecture views
  4. ISO/ IEC 29182 Part 4: Entity models
  5. ISO/ IEC 29182 Part 5: Interface definitions
  6. ISO/ IEC 29182 Part 6: Application profiles
  7. ISO/ IEC 29182 Part 7: Interoperability guidelines

4.1.2 SCADA and RFID Protocols

- IEEE created a standard specification, called Std C37.1 for SCADA and automation systems. Fig. 4.1.3 shows IEEE Std. C37.1 SCADA architecture.

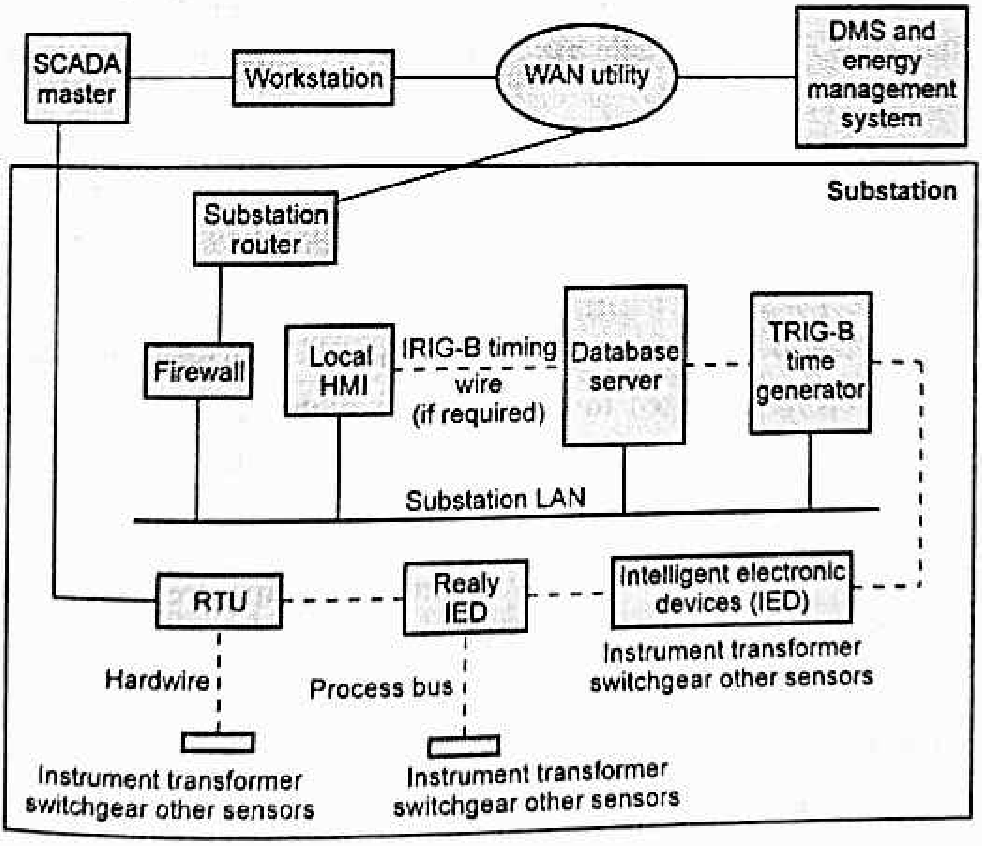

Fig. 4.1.3 : IEEE Standard C37.1 SCADA architecture

- The processing is now distributed, and functions that used to be done at the control center can now be done by the IED, that is, M2M between devices.
- **SCADA master :** This is usually a computer system (PC based systems or workstations) running the appropriate SCADA software under an appropriate operating system. These computers also provide for and interface with monitors to provide controllers with a visual state of the system.
- SCADA system consists of remote terminal units (RTUs) located in different parts of the utility, such as substations and power plants. RTUs are special purpose computers containing both analog to digital converters and digital to analog converters. They communicate their data with the SCADA master, sub-master, or with other RTUs through the multiple sports.
- Generally, RTUs interface with intelligent electronic devices (IEDs) used in transmission and distribution substations. IEDs include meters, voltage regulators, transformer tap positions, relays, re-closers and other electrical transducers.
- SCADA devices are interconnected by communication links such as optical fiber, radio, telephone line, microwave, satellite, or Ethernet. The IEEE standard C37.1-1994 specifies the communication topologies used in SCADA. They vary from simple a point-to-point to composite architectures with one single master station, multiple sub-master (slave master) stations and multiple RTUs.
- SCADA systems can also be connected to the LAN. This enables anyone within the LAN with the correct authentication and applications software to be able to access the SCADA system.
- The SCADA master also runs some selected Energy Management System (EMS) applications. RTU software is used to communicate with the master station and other RTUs. Some RTUs have sufficient processing power to be able to perform some local control.
- Other key components of the SCADA software include alarms, trends, and databases. It also needs to be fault-tolerant, scalable and able to support client/server distributed computing.
- OPC, which stands for Object Linking and Embedding (OLE) for Process Control, is the original name for a standard specification developed in 1996 by an industrial automation industry task force. OPC Clients and Servers are simply COM objects.

- OPC is an open, standards-based infrastructure for the exchange of process control data and specifies a set of software interfaces for several logical (software) objects, as well as the methods (functions) of those objects.
- The OPC Specification is a non-proprietary technical specification that defines a set of standard interfaces based upon Microsoft's OLE/COM/DCOM platform and .NET technology. The application of the OPC standard interface makes possible interoperability between automation/control applications, field systems/devices and business/office applications.
- OPC is a high-level protocol for standardizing host-to-controller communication. OPC was first designed for operation on the Microsoft Windows operating system.
- It is founded on the Component Object Model (COM) and its network distributed equivalent (DCOM). Two applications can pass messages between each other using COM and DCOM.
- COM/DCOM is based on the general remote procedure call construction and its message passing interface between executing objects is Object Linking and Embedding (OLE).
- Fig 4.1.4 shows OPC standard for device connection. OPC was designed to provide a common bridge for Windows- based software applications and process control hardware. Standards define consistent methods of accessing field data from plant floor devices.

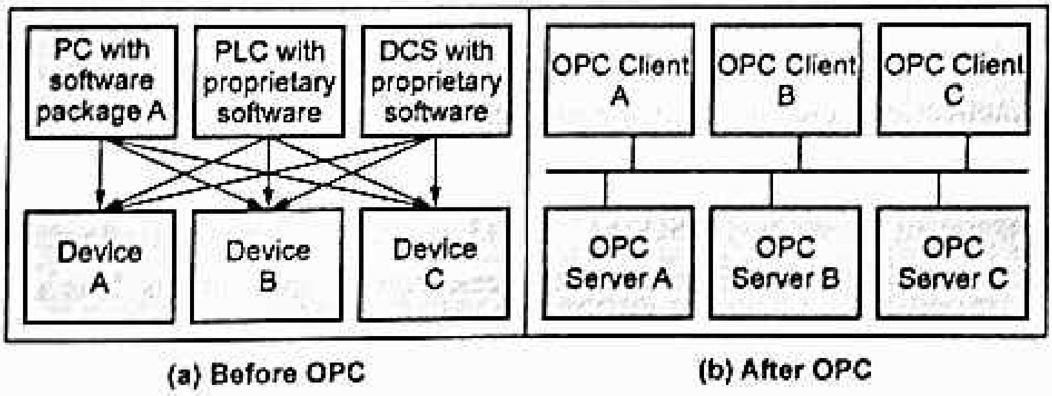

Fig. 4.1.4 : OPC standard for device connection

- Microsoft is a member of the OPC Foundation and has given strong backing to the organization. OPC provides a single, consistent, industry-standard interface that permits software suppliers to focus on adding new features to their software instead of developing long lists of proprietary hardware device drivers.
- The OPC standard has provided an environment in which device manufacturers will be encouraged to invest in the development of their own OPC servers knowing that the same server can be used by every software, HMI, PLC or DSC vendor.

- OPC enables companies to build robust and durable automation solutions quickly, choosing the right components for their specific requirements. It decreases the up-front costs and reduces long-term costs of automation and control systems. OPC Clients and Servers are simply COM objects.
- The smart cards with contactless interfaces are becoming increasingly popular for payment. Smart cards have two different types of interfaces : contact and contactless.
- Contact smart cards are inserted into a smart card reader, making physical contact with the reader. However, contactless smart cards have an antenna embedded inside the card that enables communication with the reader without physical contact. A combi card combines the two features with a very high level of security.
- Contactless smart cards contain an embedded antenna instead of contact pads attached to the chip for reading and writing information contained in the chip's memory.
- Contactless cards do not have to be inserted a smart card reader. Instead, they need only be passed within range of a radio frequency acceptor to read and store information in the chip. These cards have an antenna embedded inside the microchip that allow the card to communicate with an antenna coupler unit without physical contact.
- The standard for contactless smart card communications is ISO/ IEC 14443. The ISO/IEC 14443 is an international standard that defines the interfaces to a "close proximity" contactless smart card, including the radio frequency (RF) interface, the electrical interface, and the communications and anti-collision protocols.
- ISO/IEC 14443 compliant cards operate at 13.56 MHz and have an operational range of up to 10 centimeters (3.94 inches). ISO/IEC 14443 is the primary contactless smart card standard being used for transit, financial, and access control applications. It is also used in electronic passports and in the FIPS 201 PIV card.

4.1.3 Issues with IoT Standardization

- The rapid evolution of the IoT market has caused an explosion in the number and variety of IoT solutions. Additionally, large amounts of funding are being deployed at IoT startups. Consequently, the focus of the industry has been on manufacturing and producing the right types of hardware to enable those solutions.
- In the current model, most IoT solution providers have been building all components of the stack, from the hardware devices to the relevant cloud services or as they would like to name it as "IoT solutions", as a result, there is a lack of consistency and standards across the cloud services used by the different IoT solutions.

- Creating that model will never be an easy task by any level of imagination, there are hurdles and challenges facing the standardization and implementation of IoT solutions and that model needs to overcome all of them.
- The hurdles facing IoT standardization can be divided into four categories; Platform, Connectivity, Business Model and Applications.
  1. Platform : This part includes the form and design of the products (UI/UX), analytics tools used to deal with the massive volume of data streaming from all products in a secure way, and scalability which means that wide adoption of protocols like IPv6 in all vertical and horizontal markets is needed.
  2. Connectivity : This phase includes all parts of the consumer's day and night routine, from using wearables, smart cars, smart homes, and in the big scheme, smart cities. From the business prospective we have connectivity using Industrial Internet of Things where M2M communications dominate the field.
  3. Business Model : The bottom line is a big motivation for starting, investing in, and operating any business; without a sound and solid business model for IoT we will have another bubble, this model must satisfied all the requirements for all kinds of e-commerce; vertical markets, horizontal markets and consumer markets. But this category is always a victim of regulatory and legal scrutiny.
  4. Applications : In this category there are three functions needed to have killer applications : control "things", collect "data", and analyze "data". IoT needs killer applications to drive the business model using a unified platform.

4.2 Modbus Protocol

SPPU : May-19, Dec.-19

- Modbus is a open serial communications protocol originally published by Modicon (now Schneider Electric) in 1979 for use with its programmable logic controllers (PLCs). Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition (SCADA) systems
- MODBUS Protocol is a messaging structure developed by Modicon in 1979, used to establish master-slave/client-server communication between intelligent devices. It is a de facto standard, truly open and the most widely used network protocol in the industrial manufacturing environment.
- Modbus is used extensively in Chinese manufacturing operations. In October 2004, the Standardization Administration of China (SAC) formally launched three standards for industrial automation in the People's Republic of China :

1. GB/Z 19582.1-2004 Modbus Industrial Automation Network Specification Part 1 : Modbus Application Protocol,
2. GB/Z 19582.2-2004 Modbus Industrial Automation Network Specification Part 2 : Modbus Protocol Implementation Guide over Serial Link,
3. GB/Z 19582.3-2004 Modbus Industrial Automation Network Specification Part 3: Modbus Protocol Implementation Guide over TCP/IP.

- Modbus is not part of a physical layer on a network, as with some other protocols. Modbus communications are transferred on top of physical layers, enabling it to be utilized on many different types of networks. This non-physical layer property makes Modbus an application layer protocol.

❑ Two Variants of the Modbus Protocol

- There are two variants of the Modbus protocol that travel over serial connections : RTU and ASCII.

1. **RTU :** This variation is more compact, and uses binary communication. In this format, data transmissions are always followed by a cyclic redundancy check checksum, which are used to detect transmission problems.
2. **ASCII :** This version is more verbose, and it uses hexadecimal ASCII encoding of data that can be read by human operators. A different type of checksum, the longitudinal redundancy check checksum, takes place after Modbus ASCII data transmissions. Modbus ASCII is the less secure of the two variants.

❑ Modbus is a "Master/Slave " Protocol

- Modbus communications take place between a centralized master and up to 247 connected electronic devices on a single network. The design is commonly referred to as a "master/slave" protocol, because the system "master" requests information from connected devices, which are referred to as "slaves".
- Slave devices only send information to the master in response to these requests, and do not operate autonomously. The master can also write information to the slave devices, but the slave devices cannot write information to the master.
- When a slave device transmits a communication to the Modbus master, it begins the message with a unique address identifier. This is a number ranging from 1 to 247. This enables the master to identify which specific device is responding with the requested information.

❑ MODBUS MESSAGE FRAMING

- A message frame is used to mark the beginning and ending point of a message allowing the receiving device to determine which device is being addressed and to know when the message is completed. It also allows partial messages to be detected and errors flagged as a result.
- A MODBUS message is placed in a message frame by the transmitting device. Each word of this message (including the frame) is also placed in a data frame that appends a start bit, stop bit, and parity bit.
- In ASCII mode, the word size is 7 bits, while in RTU mode; the word size is 8 bits. Thus, every 8 bits of an RTU message is effectively 11 bits when accounting for the start, stop, and parity bits of the data frame

❑ Modbus Applications

- Modbus is used to communicate between intelligent devices and sensors and instruments, and to monitor field devices using PCs and human-machine interfaces.
- Modbus is most widely used as an industrial protocol, but is also popular in building, infrastructure, transportation, and energy applications

Review Questions

1. *Explain the Modbus message framing and transmission modes.* **SPPU : May-19, Marks 6**
2. *Explain Mod Bus protocol in detail.* **SPPU : Dec.-19, Marks 7**

4.3 Zigbee Architecture

SPPU : May-18, 19, Dec.-18

- ZigBee is built on top of the IEEE 802.15.4 standard. ZigBee provides routing and multi-hop functions to the packet-based radio protocol. ZigBee is a registered trademark of the ZigBee Alliance. 802.15.4™ is a trademark of the Institute of Electrical and Electronics Engineers (IEEE). 802.15.4 defines the physical and MAC layers, and ZigBee defines the network and application layers.
- ZigBee is the primary protocol which builds on the 802.15.4 standard, adding a network layer capable of peer-to-peer multi-hop mesh networking, a security layer capable of handling complex security situations, and an application layer for interoperable application profiles.

- The 802.15.4 specification was created and is maintained by IEEE. This specification defines the physical (PHY) and media access control (MAC) layers of a personal area, low-power, wireless network
- ZigBee is a wireless standard for home and commercial use developed by the ZigBee Alliance, established in 2002. It is based on an IEEE 802.15.4 standard. The latest version of the standard is known as ZigBee Pro and was published in 2007.
- A major feature of the ZigBee protocol is its mesh network topology that is self-healing and auto-routing. Mesh networks do not depend on any single connection; if one link is broken, devices search through the mesh to find another available route. This capability makes a ZigBee-based network very reliable and flexible.
- The protocol is administered by the ZigBee Alliance, an open, non-profit association of approximately 400 members. The Alliance certifies products and promotes worldwide adoption of ZigBee as the wirelessly networked standard for sensing and control in consumer, commercial and industrial areas.
- The ZigBee specifications enhance the IEEE 802.15.4 standard by adding network and security layers and an application framework.
- ZigBee is an open, global, packet-based protocol designed to provide an easy-to-use architecture for secure, reliable, low power wireless networks. Flow or process control equipment can be place anywhere and still communicate with the rest of the system. It can also be moved, since the network doesn't care about the physical location of a sensor, pump or valve.
- ZigBee targets the application domain of low power, low duty cycle and low data rate requirement devices. Figure below shows the example of a ZigBee network.
- IEEE 802.15.4 supports star and peer-to-peer topologies. The ZigBee specification supports star and two kinds of peer-to-peer topologies, mesh and cluster tree. ZigBee-compliant devices are sometimes specified as supporting point-to-point and point-to-multipoint topologies

❑ Features

1. Stochastic addressing : A device is assigned a random address and announced. Mechanism for address conflict resolution. Parents node don't need to maintain assigned address table.
2. Link Management : Each node maintains quality of links to neighbors. Link quality is used as link cost in routing.

3. Frequency Agility : Nodes experience interference report to channel manager, which then selects another channel
4. Asymmetric Link : Each node has different transmit power and sensitivity. Paths may be asymmetric.
5. Power Management : Routers and Coordinators use main power. End Devices use batteries.

4.3.1 ZigBee Architecture

- Fig. 4.3.1 shows position of ZigBee and IEEE 802.15.4.
- "ZigBee specifies all the layers above MAC and PHY, including the NWK (network) layer, APS, ZDO, and security layers. It's ZigBee that provides the mesh networking and multi-hop capabilities, enhances the reliability of data packet delivery, and specifies application-to-application interoperability.
- ZigBee does not use all of the 802.15.4 MAC/PHY specification; only a small subset. This allows stack vendors to offer smaller solutions by providing a limited MAC for their ZigBee stacks.
- ZigBee is asynchronous. Any node may transmit at any time. Only the CSMA/CA , a MAC-level mechanism, prevents nodes from talking over each other.
- Application (APL) Layer : The top layer in the ZigBee protocol stack consists of the Application Framework, ZigBee Device Object (ZDO), and Application Support (APS) Sublayer.

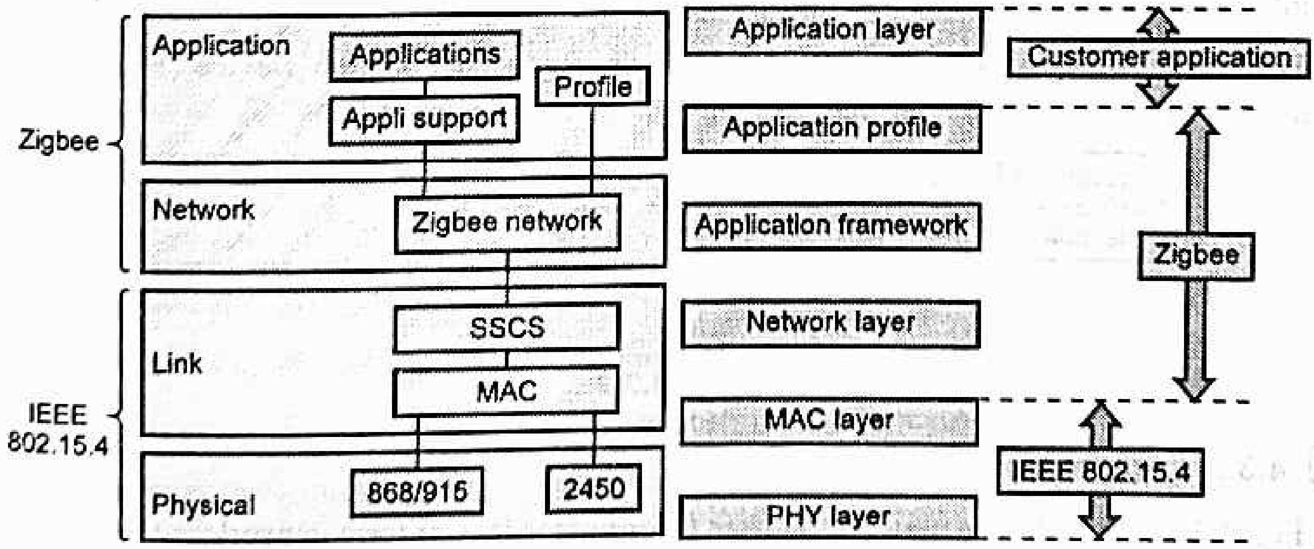

Fig. 4.3.1 : Zigbee stack

- Application Framework : It provides a description of how to build a profile onto the ZigBee stack. It also specifies a range of standard data types for profiles, descriptors to assist in service discovery, frame formats for transporting data, and a key value pair construct to rapidly develop simple attribute-based profiles.
- Software at an endpoint that controls the ZigBee device. A single ZigBee node supports up to 240 application objects. Each application object supports endpoints numbered between 1 and 240 (with endpoint 0 reserved for the ZigBee Device Object (ZDO).
- ZigBee Cluster Library (ZCL) is a library of clusters that can be used by any application. This allows common clusters to be reused across a number of different functional domains, for example, the same lighting clusters can be used for any application that requires lighting controls, such as home automation and commercial building automation.
- Each cluster has two "ends" , client and server.

❑ Working of Zigbee

- ZigBee basically uses digital radios to allow devices to communicate with one another. A typical ZigBee network consists of several types of devices. A network coordinator is a device that sets up the network, is aware of all the nodes within its network, and manages both the information about each node as well as the information that is being transmitted/received within the network.
- Every ZigBee network must contain a network coordinator. Other Full Function Devices (FFD's) may be found in the network, and these devices support all of the 802.15.4 functions. They can serve as network coordinators, network routers, or as devices that interact with the physical world.
- The final device found in these networks is the Reduced Function Device (RFD), which usually only serve as devices that interact with the physical world.

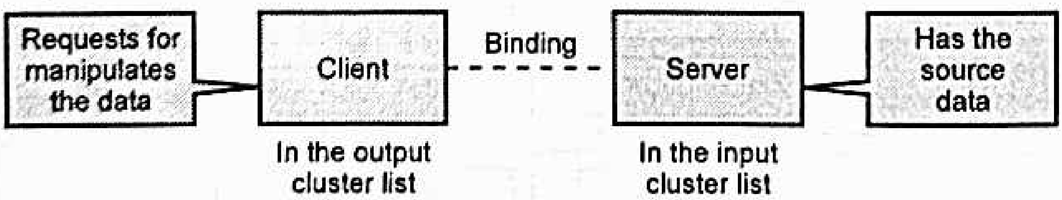

Fig. 4.3.2

4.3.2 Logical Device Types

- Fig. 4.3.3 shows Zigbee network. ZigBee Coordinators (ZCs) form networks.
- The ZigBee stack resides on a ZigBee logical device. There are three logical device types : Coordinator, Router and End device

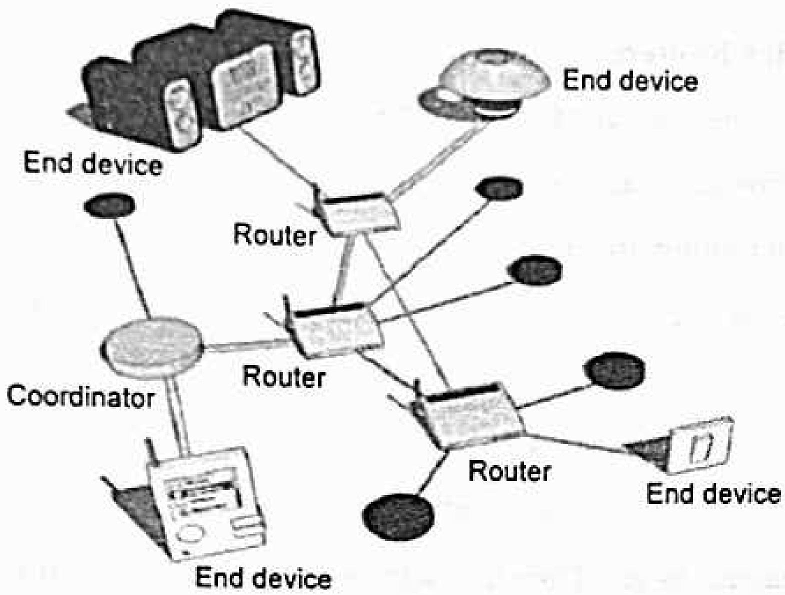

Fig. 4.3.3 : Zigbee network

- It is at the network layer that the differences in functionality among the devices are determined. It is expected that in a ZigBee network the coordinator and the routers will be mains-powered and that the end devices can be battery-powered.
- In a ZigBee network there is one and only one coordinator per network. The number of routers and/or end devices depends on the application requirements and the conditions of the physical site.
- Within networks that support sleeping end devices, the coordinator or one of the routers must be designated as a Primary Discovery Cache Device. These cache devices provide server services to upload and store discovery information, as well as respond to discovery requests, on behalf of the sleeping end devices.
- **Function of ZigBee Coordinator :**
  1. It forms a network.
  2. It establishes the 802.15.4 channel on which the network will operate.
  3. It establishes the extended and short PAN ID for the network.
  4. It decides on the stack profile to use (compile or run-time option).
  5. It acts as the Trust Center for secure applications and networks.
  6. It acts as the arbiter for End-Device-Bind (a commissioning option).
  7. It acts as a router for mesh routing.
  8. It acts as the top of the tree, if tree routing enabled.

- **Function of ZigBee Routers :**
  1. Finding and joining the " correct " network
  2. Perpetuating broadcasts across the network
  3. Participating in routing, including discovering and maintaining routes
  4. Allowing other devices to join the network (if permit-join enabled)
  5. Storing packets on behalf of sleeping children
- **Functions of ZigBee End-Devices :**
  1. Finding and joining the " correct " network
  2. Polling their parents to see if any messages were sent to them while they were asleep
  3. Finding a new parent if the link to the old parent is lost (NWK rejoin)
  4. Sleeping most of the time to conserve batteries when not in use by the application.

ZigBee Network Formation

1. Forming a Network

- The coordinator initiates network formation.
- After forming the network, the coordinator can function as a router & can accept requests from other devices wishing to join the network.
- Depending on the stack and application profile used, the coordinator might also perform additional duties after network formation.

2. Joining a Network

- A device finds a network by scanning channels.
- When a device finds a network with the correct stack profile that is open to joining, it can request to join that network.
- A device sends its join request to one of the network's router nodes & the device receiving the request can then use the callback to accept the request or deny it.

3. Network Communication

- All nodes that communicate on a network transmit and receive on the same channel, or frequency.
- ZigBee uses a personal area network identifier (PAN ID) to identify a network this provides a way for two networks to exist on the same channel while still maintaining separate traffic flow.

4.3.3 Network Layer

- The Zigbee networking layer is probably one of the most complex layers in the protocol stack.
- Network layer responsibilities :
  1. Establishment of a new network.
  2. New device configuration, addressing assignment, network synchronization
  3. Frames security
  4. Message routing.
- On the transmit side, the Application Sub Layer (APS) sends data down to the network layer via a data request. Inside the data request, the APS specifies the destination address, whether or not route discovery will be allowed, and the radius.
- The radius is the maximum hops that the frame will be allowed to travel. This is mostly used in broadcasts. Once inside the data request, the network header is filled out and then the address is examined.
- At this point, it's probably best to make a distinction between network addressing and MAC addressing. There are two sets of addresses in each Zigbee frame : the 802.15.4 source/destination addresses and the Zigbee network source/destination addresses.
- ZigBee uses two unique addresses per node : a long address (MAC address) and a short address (Network address).

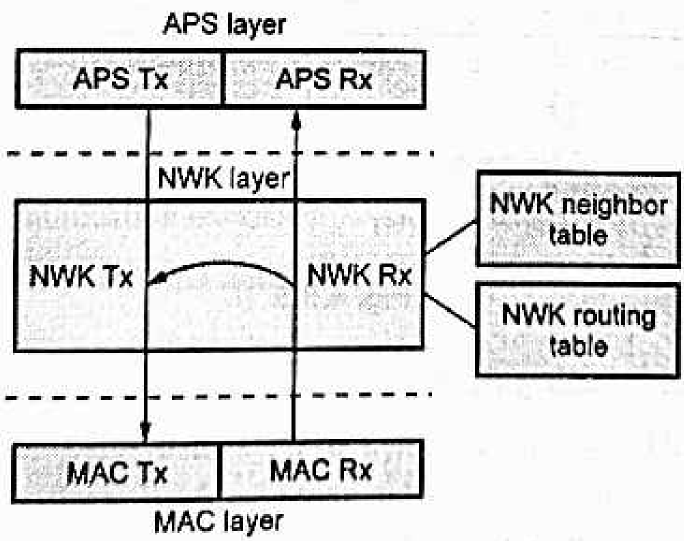

Fig. 4.3.4

- The long address, also called the IEEE or MAC address, is assigned by the manufacturer of a device using an 802.15.4 radio), and does not change for the life of the device. The long address uniquely identifies the widget from all other 802.15.4 widgets in the world. In fact, the 64 bits (8 bytes) of this address space are large enough to include 123,853 devices for every square meter on earth.
- The IEEE address space, surprisingly enough, is controlled by IEEE. A block of them can be purchased by the manufacturer. Each IEEE address uniquely identifies this node from any other node in the world. The top three bytes (24 bits) are called the Organizational Unique Identifier (OUI), and the bottom five bytes (40 bits) are managed by the Original Equipment Manufacturer (OEM).

❑ Components :

1. Network Layer Data Entity (NLDE) : Makes NPDU from NSDU
2. Network Layer Management Entity (NLME) : Configure new device, neighbor discovery, route discovery, joining/leaving a network, ...
3. Network Layer Information Base (NIB) : Capabilities (RFD/FFD), Security level, protocol version, route discovery time, max retries for route discovery, neighbor table,

- Fig. 4.3.5 ZigBee Network Layer Frame Format.

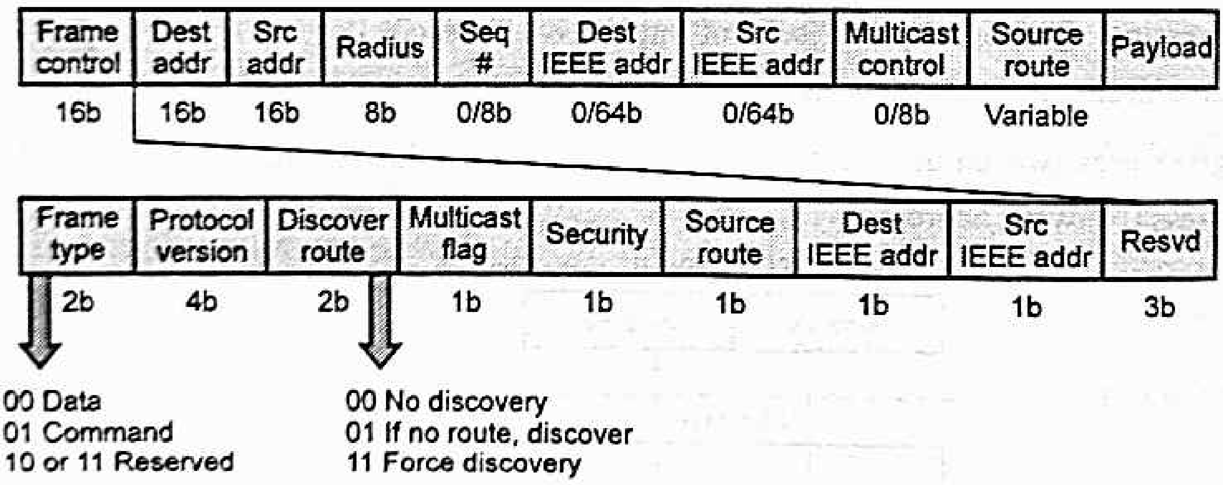

Fig. 4.3.5

4.3.4 Simple Tree Routing Protocol

- The tree network addresses are assigned using a distributed addressing scheme. These addresses are unique within a particular network. As a result, a router can acquire the relationship of two nodes through their network addresses.
- The ZC becomes the root of the tree. Every device except the ZC has a parent node, while a ZR may also has one or more child nodes. A ZED acts as a leaf in the tree topology.

- Fig. 4.3.6 shows tree architecture.

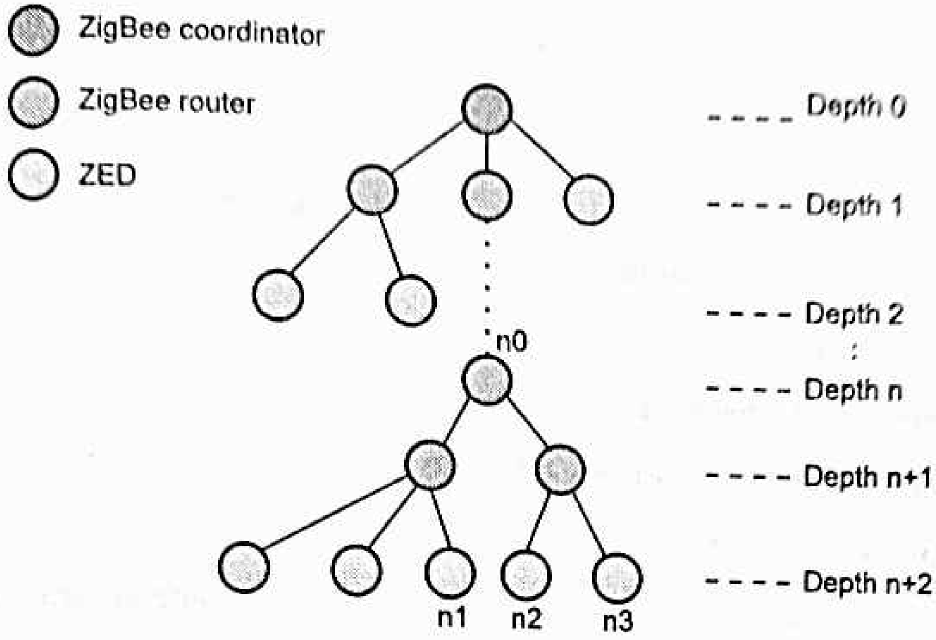

Fig. 4.3.6 : Tree architecture

- In a tree topology, the role of the ZigBee coordinator is to establish the network and configure all parameters. During the establishment phase, the ZigBee coordinator determines the maximum number of nodes, which are (Cm) and (Rm).
- Whereas Cm is the maximum number of children and 'Rm' is the maximum number of routers a parent may have as children. In addition, each node has an attribute called "depth" which is the minimum number of hops to reach the coordinator using only parent-child link. The ZigBee coordinator itself has a depth of zero and it determines the maximum tree depth of the network (Lm).
- Tree routing, also acknowledged, is only available in Stack Profile 0x01. Tree is just as bandwidth-efficient as mesh, and more efficient in terms of memory. But it has a major drawback: If links between parent and children break, it has no recovery. So ZigBee uses mesh by default.
- **Advantages :** No communication overhead
- **Disadvantages :** More transmission delay (not optimal path) and Single point of failure

4.3.5 ZigBee-AODV (ZAODV)

- Another routing protocol in ZigBee is a route discovery protocol similar to Ad-hoc On-demand Distance Vector Routing.
- AODV is a reactive protocol, i.e., the network stays silent until a connection is requested. If a node wants to communicate with another one, it broadcasts a request to its neighbors who re-route the message and safeguard the node from which they received the message.

- If a node receives a message and it has an entry corresponding to the destination in its routing table, it returns a RREP through the reverse-path to the requesting node. So, the source sends its data through this path to the destination with the minimum number of hops
- In ZAODV, when a node desires to send data to another node without having the related routing information, it buffers the data and then initiates a route discovery process to find the optimal path for the destination.
  1. **Routing Table(RT)**
  2. **Route Discovery Table(RDT)**
  3. **Route Request Command Frame (RREQ)**
  4. **Route Reply Command Frame(RREP)**
- The ZigBee routing algorithm uses a path cost metric for route comparison during route discovery and maintenance.

❑ RREQ

1. The source node broadcasts a route request (RREQ) packet.
2. If a intermediate router receives an new RREQ, it add RDT entry and rebroadcast the RREQ.
3. If a intermediate router receives an old RREQ, it shall compare the Forward Cost in the RREQ to the one stored in the RDT entry.
4. If the Cost is less, the router shall update the RDT entry and rebroadcast the RREQ again.
5. Using Source Address and Route Request ID, the RREQ can be uniquely identified.

❑ RREP

1. The destination node receives the RREQ, it responds by unicasting a route reply (RREP) packet to its neighbor from which it received the RREQ.
2. The destination will choose the routing path with the lowest cost and then send a route reply.
3. Finally, when the RREP reaches the originator, the route discovery process is finished.

- ZAODV can effectively reduce the transmission delay. However, the procedure of flooding RREQ wastes significant amount of bandwidth, memory and device energy, and might cause a "broadcast storm" problem.

4.3.6 APS Layers

- Application Support Sublayer (APS) is responsible for :
  1. binding tables
  2. message forwarding between bound devices
  3. group address definition and management
  4. address mapping from 64-bit extended addresses to 16-bit NWK addresses
  5. fragmentation and reassembly of packets
  6. reliable data transport
- The key to interfacing devices at the need/service level is the concept of binding. Binding tables are kept by the coordinator and all routers in the network. The binding table maps a source address and source endpoint to one or more destination addresses and endpoints. The cluster ID for a bound set of devices will be the same.
- APS layer is responsible for providing a data service to the application and ZigBee device profiles. It also provides a management service to maintain binding links and the storage of the binding table itself.
- ZigBee's security services include methods for key establishment and transport, device management, and frame protection. The ZigBee specification defines security for the MAC, network and APS layers. Security for applications is typically provided through Application Profiles.
- The APS layer allows frame security to be based on link keys or the network key. Encryption can be applied at three different levels, the MAC layer, Network layer and the Application Support (APS) layer. A ZigBee frame will contain fields from all of these layers encapsulated within one another, and each later can encrypt its data payload.
- The APS layer is also responsible for providing applications and the ZDO with key establishment, key transport, and device management services. Fig 4.3.7 shows Zigbee frame with security on The APS level.

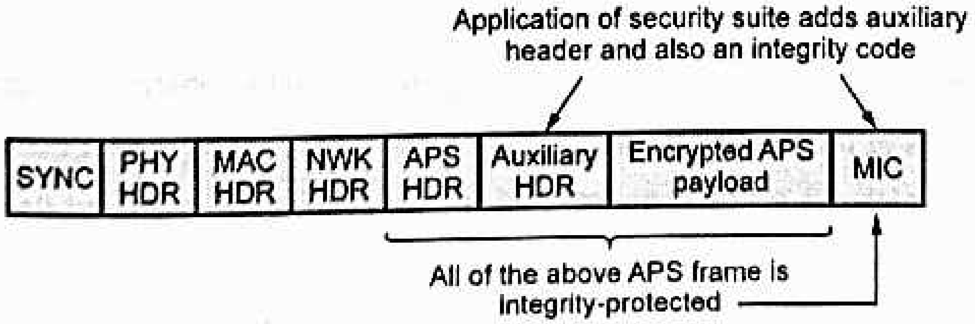

Fig. 4.3.7 : Zigbee frame with security on the APS level

- The APS sublayer is key establishment services provide the mechanism by which a ZigBee device may derive a shared secret key, with another ZigBee device. Key establishment involves two entities, an initiator device and a responder device, and is prefaced by a trust-provisioning step.

4.3.7 ZigBeel Applications

- There are numerous applications that are ideal for the redundant, self-configuring and self-healing capabilities of ZigBee wireless mesh networks. Key ones include :

  1. Energy Management and Efficiency :-To provide greater information and control of energy usage, provide customers with better service and more choice, better manage resources, and help to reduce environmental impact.
  2. Home Automation :- To provide more flexible management of lighting, heating and cooling, security, and home entertainment systems from anywhere in the home.
  3. Building Automation :- To integrate and centralize management of lighting, heating, cooling and security.
  4. Industrial Automation :- To extend existing manufacturing and process control systems reliability.

Review Questions

1. *Explain the Zigbee architecture with suitable diagram.* **SPPU : May-18, 19, Marks 6**
2. *Explain ZigBee protocol in detail.* **SPPU : Dec.-18, Marks 7**
3. *Explain the Zigbee node types and their functions.* **SPPU : May-19, Marks 6**

4.4 IP based Protocols

- Internet Protocol (IP) provide delivery of packets from one host in the Internet to any other host in the Internet, even if the hosts are on different networks.
- It provides unreliable and connectionless datagram delivery service
- Internet packets are called "datagrams" and may be up to 64 kilobytes in length.
- Key advantages of Internet Protocol :

  1. Versatile : Layered IP architecture is well equipped to cope with any of physical and data link layers.
  2. Ubiquitous : All recent OS releases have an integrated dual stack that gets enhanced over time.

3. Scalable : IP has massively deployed and tested for robust scalability.
4. IP is manageable and highly secure.
5. It is open technology and based on standards.
6. It is stable and resilient.

4.4.1 MQTT

- Message Queue Telemetry Transport (MQTT) is Open Connectivity for Mobile, M2M and IoT.
- MQTT is designed for high latency, low-bandwidth or unreliable networks. The design principle minimizes the network bandwidth and device resource requirements.
- MQTT is a lightweight broker-based publish/subscribe messaging protocol designed to be open, simple, lightweight and easy to implement.

MQTT characteristics

1. Lightweight message queueing and transport protocol
2. Asynchronous communication model with messages (events)
3. Low overhead (2 bytes header) for low network bandwidth applications
4. Publish / Subscribe (PubSub) model
5. Decoupling of data producer (publisher) and data consumer (subscriber) through topics (message queues)
6. Simple protocol, aimed at low complexity , low power and low footprint implementations
7. Runs on connection-oriented transport (TCP).
8. MQTT caters for (wireless) network disruptions

- The MQTT protocol works by exchanging a series of MQTT control packets in a defined way. Each control packet has a specific purpose and every bit in the packet is carefully crafted to reduce the data transmitted over the network.
- A MQTT topology has a MQTT server and a MQTT client. MQTT control packet headers are kept as small as possible.
- Having a small header overhead makes this protocol appropriate for IoT by lowering the amount of data transmitted over constrained networks.
- MQTT is the protocol built for M2M and IoT which is used to provide new and revolutionary performance.

- Fig. 4.4.1 shows MQTT publish/subscribe framework.

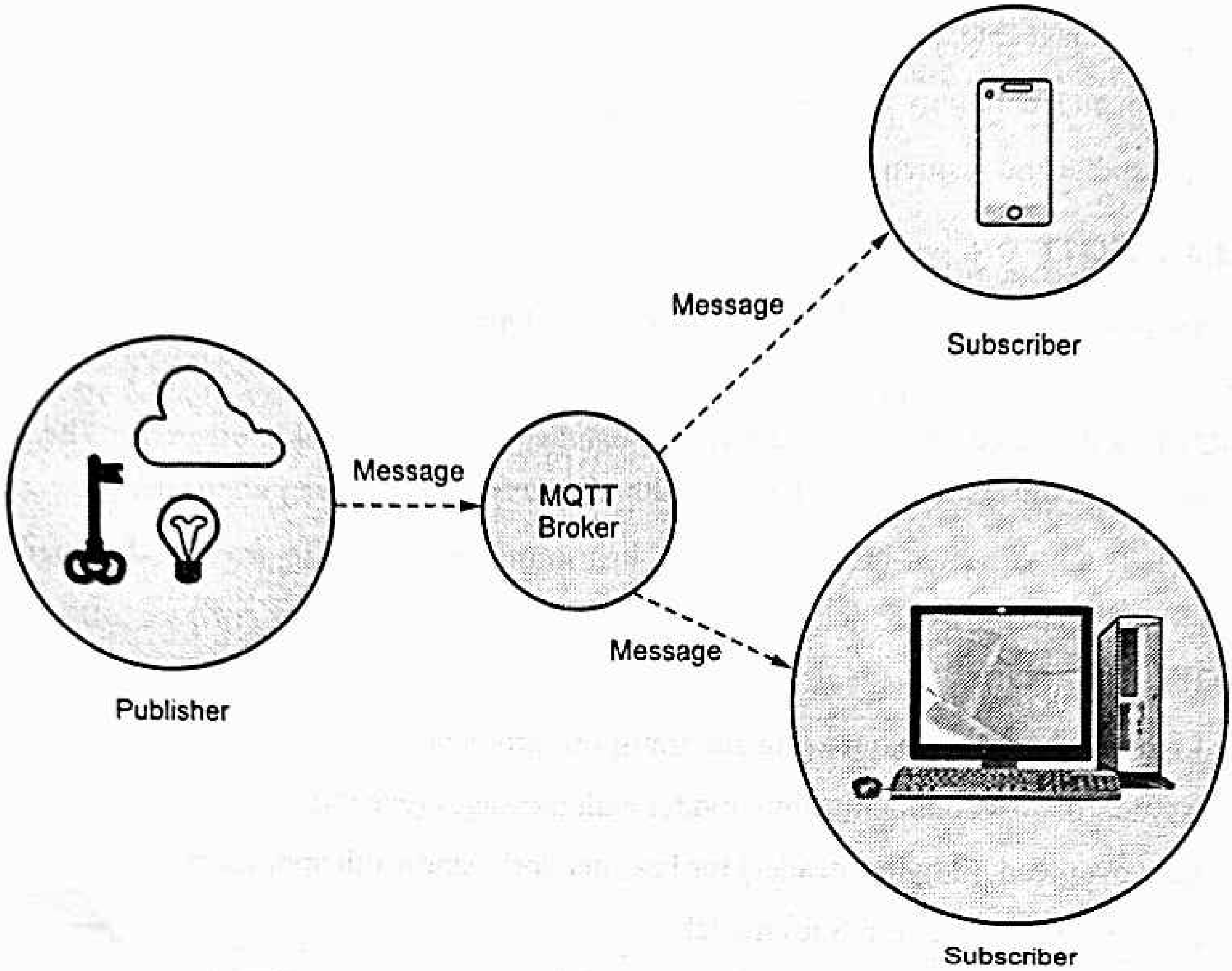

Fig. 4.4.1 : MQTT publish/subscribe framework

- A producer publishes a message (publication) on a topic (subject). A consumer subscribes (makes a subscription) for messages on a topic (subject).
- A message server (called BROKER) matches publications to subscriptions.
- If none of them match the message is discarded after modifying the topic. If one or more matches the message is delivered to each matching consumer after modifying the topic
- Publish / Subscribe has three important characteristics :
  1. It decouples message senders and receivers, allowing for more flexible applications.
  2. It can take a single message and distribute it to many consumers.
  3. This collection of consumers can change over time, and vary based on the nature of the message.
- The MQTT messages are delivered asynchronously ("push") through publish subscribe architecture.

- The MQTT protocol works by exchanging a series of MQTT control packets in a defined way.
- Each control packet has a specific purpose and every bit in the packet is carefully crafted to reduce the data transmitted over the network.
- A MQTT topology has a MQTT server and a MQTT client. MQTT client and server communicate through different control packets. Table below briefly describes each of these control packets
- MQTT control packet headers are kept as small as possible. Each MQTT control packet consist of three parts, a fixed header, variable header and payload.
- Each MQTT control packet has a 2 byte Fixed header. Not all the control packet have the variable headers and payload.
- A variable header contains the packet identifier if used by the control packet. A payload up to 256 MB could be attached in the packets.
- Having a small header overhead makes this protocol appropriate for IoT by lowering the amount of data transmitted over constrained networks.

❑ MQTT Quality of Service :

- There are the three levels of MQTT QoS.

1. **QoS 0 : AT MOST ONCE**
   - Guarantees that a particular message is only ever received by the subscriber a maximum of one time. This does mean that the message may never arrive.
   - The sender and the receiver will attempt to deliver the message, but if something fails and the message does not reach its destination the message may be lost.
   - This QoS has the least network traffic overhead and the least burden on the client and the broker and is often useful for telemetry data where it doesn't matter if some of the data is lost.
2. **QoS 1 : AT LEAST ONCE**
   - Guarantees that a message will reach its intended recipient one or more times. The sender will continue to send the message until it receives an acknowledgment from the recipient, confirming it has received the message.
   - The result of this QoS is that the recipient may receive the message multiple times, and also increases the network overhead than QoS 0.
   - In addition more burden is placed on the sender as it needs to store the message and retry should it fail to receive an ack in a reasonable time.

3. **QoS 2 : EXACTLY ONCE**

- The most costly of the QoS, this QoS will ensure that the message is received by a recipient exactly one time.
- This ensures that the receiver never gets any duplicate copies of the message and will eventually get it, but at the extra cost of network overhead and complexity required on the sender and receiver.

4.4.2 Difference between CoAP and MQTT

CoAP	MQTT
CoAP uses UDP protocol.	MQTT uses TCP protocol.
It uses Request/Response messaging.	It uses publish/subscribe messaging.
Communication model is One-to-one.	Communication model is Many-to-many.
Advantages : • Lightweight and fast • Low overhead • Support for multicasting	**Advantages :** • Simple management • Scalability • Robust communication
Weakness : Not as reliable as TCP based MQTT	Weakness : Higher overhead, no multicasting support
Security type is DTLS.	Security type is SSL/TLS.
Effectiveness in LLN is excellent.	Effectiveness in LLN is low.

4.4.3 6LoWPAN

- In order to enable IPv6 over IEEE 802.15.4 networks, the adaptation layer called 6LoWPAN was developed.
- Subsequently, 6Lo made use of 6LoWPAN as a basis to support IPv6 over Bluetooth LE, ITU-T G.9959, DECT ULE, MS/TP, NFC, IEEE 1901.2, and IEEE 802.11ah.
- A key IP based technology is 6LowPAN (IPv6 Low-power wireless Personal Area Network). Rather than being an IoT application protocols technology like Bluetooth or ZigBee, 6LowPAN is a network protocol that defines encapsulation and header compression mechanisms.
- 6LoWPAN provides a means of carrying packet data in the form of IPv6 over IEEE 802.15.4 and other networks. It provides end-to-end IPv6 and as such it is able to provide direct connectivity to a huge variety of networks including direct connectivity to the Internet.

- In order to send packet data, IPv6 over 6LowPAN, it is necessary to have a method of converting the packet data into a format that can be handled by the IEEE 802.15.4 lower layer system.
- IPv6 requires the maximum transmission unit (MTU) to be at least 1280 bytes in length. This is considerably longer than the IEEE802.15.4's standard packet size of 127 octets which was set to keep transmissions short and thereby reduce power consumption.
- To overcome the address resolution issue, IPv6 nodes are given 128-bit addresses in a hierarchical manner. The IEEE 802.15.4 devices may use either of IEEE 64-bit extended addresses or 16-bit addresses that are unique within a PAN after devices have associated. There is also a PAN-ID for a group of physically co-located IEEE802.15.4 devices.
- Fig. 4.4.2 shows the 6LoWPAN protocol stack compared to the IP protocol stack.

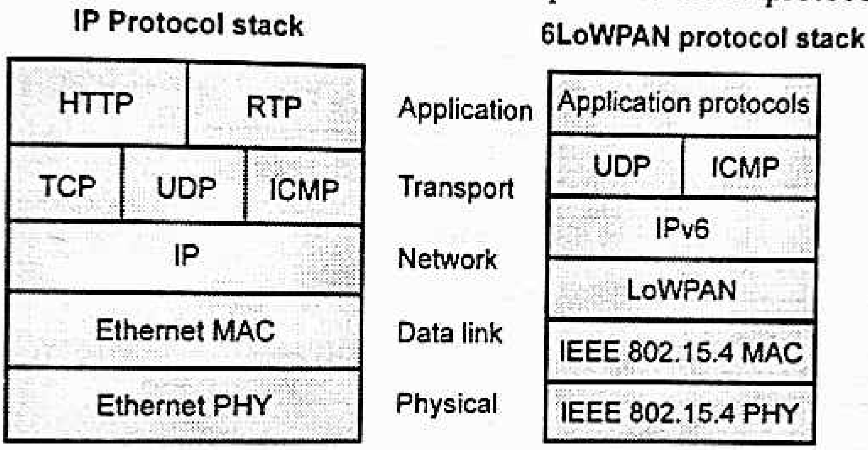

Fig. 4.4.2 : 6LoWPAN protocol stack compared to the IP protocol stack

- It is almost identical to a normal IPv6 implementation with the following two differences: x 6LoWPAN only supports IPv6, for which a small adaptation layer (LoWPAN) has been defined to optimize IPv6 over link layers. x Although 6LoWPAN is not bound to the IEEE 802.15.4 standard, it is designed to utilize it.
- Fig. 4.4.3 shows 6LoWLAN header stack.

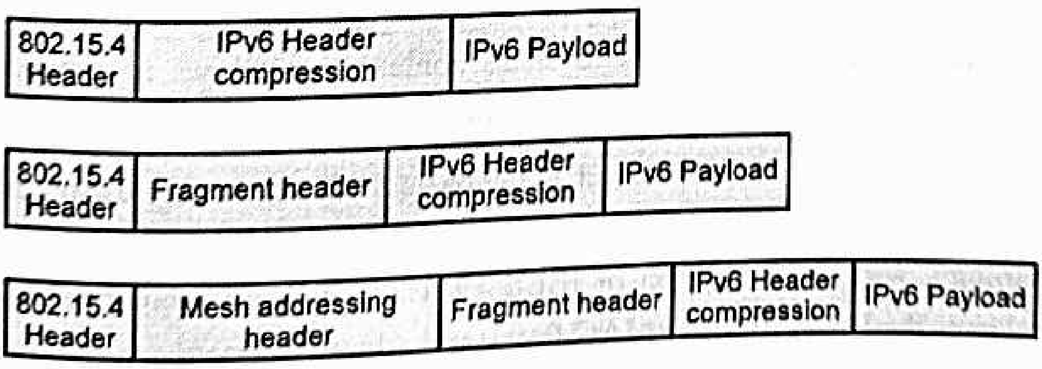

Fig. 4.4.3 : 6LoWLAN header stack

❑ Header compression

- 6LoWPAN uses header compression mechanisms to reduce the packet overhead when sending data over 802.15.4 links. It effectively reduces the size of the IPv6 header and transport headers sent over-the-air by relying on a set of assumptions and information present in the link layer
- When communicating an IPv6 packet over IEEE 802.15.4 or a 6Lo technology, a typical 40-byte IPv6 header would consume a significant fraction of frame payload, as well as precious energy and bandwidth resources.
- 6LoWPAN header compression was defined to produce a lightweight encoding for the IPv6 header over IEEE 802.15.4. 6LoWPAN HC leverages stateless and stateful techniques.
- Fig. 4.4.4 shows header compression.

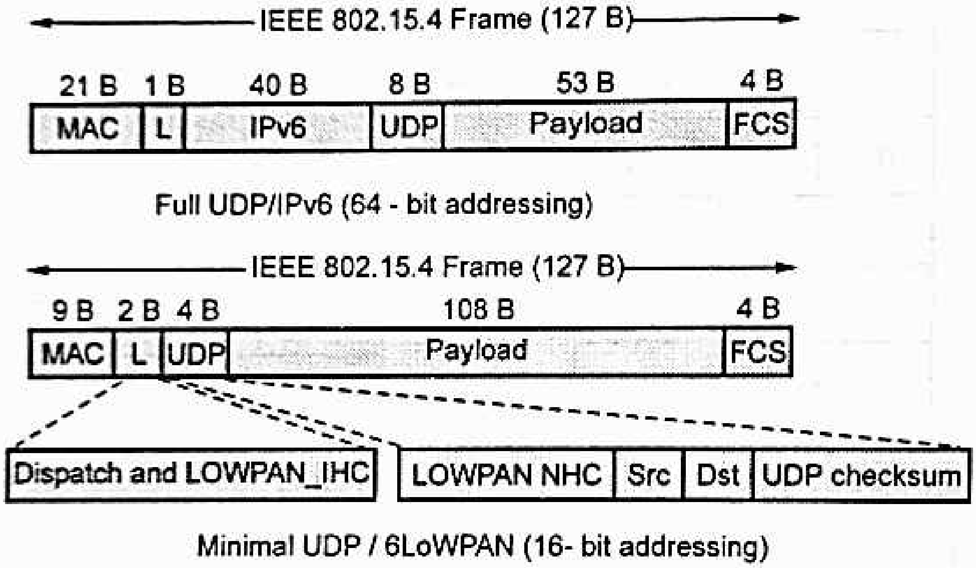

Fig. 4.4.4 : Header compression

- The former exploits the fact that some IPv6 header fields (e.g., IIDs, and datagram length) are derivable from the link layer header, and an optimistic expectation that several IPv6 header fields will be set to typical values.
- The latter requires 6LoWPAN nodes to share context that allows to substitute address prefixes or complete addresses by a short identifier.
- Subsequently, 6LoWPAN HC has served as the basis for header compression over the 6Lo technologies. 6LoWPAN HC has needed some tailoring to the specific characteristics of each technology, with the exception of IEEE 1901.2, which does not need any changes, since it endorses the IEEE 802.15.4 MAC frame and address formats.

❑ Fragmentation :

- Fragment header composed by three primary fields: datagram size, datagram tag and datagram offset.
- Fig. 4.4.5 shows 6LoWPAN fragmentation header.

802.15.4 Header	6LoWPAN fragmentation header	Datagram size	Datagram tag	Datagram offset	FCS

Fig. 4.4.5

- The data link of IEEE 802.15.4 with a frame length of maximum 127 bytes does not match the MTU of IPv6, which is 1280 bytes.
- Datagram size describes the total (un-fragmented) payload. Datagram tag identifies the set of fragments and is used to match fragments of the same payload.
- Datagram offset identifies the fragment's offset within the un-fragmented payload. The fragment header length is 4 bytes for the first header and 5 bytes for all subsequent headers

❑ Mesh Addressing

- The mesh address header is used to forward packets of multiple hops inside a 6LoWPAN network.
- The mesh address header includes three fields: hop limit, source address and destination address.
- The hop limit field is used to limit the number of hops for forwarding. The field is decremented at each hop. Once the count reaches zero the packet is dropped.
- The source and destination address fields indicate the IP end points.
- Both are IEEE 802.15.4 addresses and may be short or extended as defined in the IEEE 802.15.4 standard. The mesh address header's length is between 5 and 17 bytes, depending on the addressing mode in use

❑ Routing

- Routing is the ability to send a data packet from one device to another device, sometimes over multiple hops.
- Depending on what layer the routing mechanism is located, two categories of routing are defined: mesh-under or route-over.

- Mesh-under uses the layer-two (link layer) addresses (IEEE 802.15.4 MAC or short address) to forward data packets; while route-over uses layer three (network layer) addresses (IP addresses).
- In a mesh-under system, routing of data happens transparently, hence mesh-under networks are considered to be one IP subnet. The only IP router in such a system is the edge router.
- One broadcast domain is established to ensure compatibility with higher layer IPv6 protocols such as duplicate address detection.
- These messages have to be sent to all devices in the network, resulting in high network load. Mesh-under networks are best suited for smaller and local networks.
- In route-over networks the routing takes place at the IP level, thus each hop in such networks represents one IP router.
- The usage of IP routing provides the foundation to larger and more powerful and scalable networks, since every router must implement all features supported by a normal IP router such as DAD, etc.
- The most widely used routing protocol for route-over 6LoWPAN networks today is RPL as defined by IETF in RFC 6550.
- Compared to mesh-under, route-over features the advantage that most of the protocols used on a standard TCP/IP stack today can be implemented and used as is.

4.4.4 LoRa

- LoRa is Long Range, low data rate, low power wireless platform technology for building IoT network.LoRa is a patented digital wireless data communication IoT technology developed by Cycleo of Grenoble, France.
- LoRa transmits over license-free Megahertz radio frequency bands like 169 MHz, 433 MHz, 868 MHz (Europe) and 915 MHz (North America). LoRa enables very-long-range transmissions (more than up to 10 miles in rural areas) with low power consumption.
- LoRa provides for long-range communications: up to three miles (five kilometers) in urban areas, and up to 10 miles (15 kilometers) or more in rural areas (line of sight). A key characteristic of the LoRa-based solutions is ultra-low power requirements, which allows for the creation of battery-operated devices that can last for up to 10 years.
- The LoRa Alliance developed the LoRaWAN protocol for use by mobile network operators who want to use unlicensed spectrum to communicate with IoT devices in their network.

- LoRaWAN defines the communication protocol and system architecture for the network, while the LoRa physical layer enables the long-range communication link. LoRaWAN communication protocol ensures reliable communication and secure communication.
- Fig. 4.4.6 shows LoRa network.

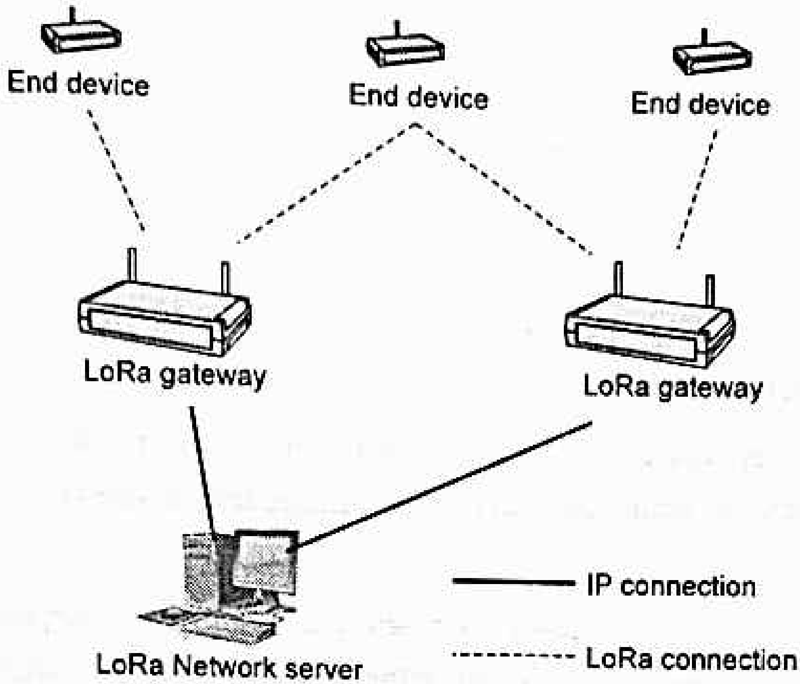

Fig. 4.4.6 : LoRa network

- LoRa network uses a star topology in which an end node can send messages to multiple gateways that communicate with the network server. Since an end node does not belong to a specific gateway, more than one gateway can receive a message sent by an end device.
- LoRa radio access technology is used in communications between an end device and the gateways. The gateways and network server are connected via standard IP connections.
- The LoRa sensors transmit data to the LoRa gateways. The LoRa gateways connect to the internet via the standard IP protocol and transmit the data received from the LoRa embedded sensors to the Internet i.e. a network, server, or cloud.
- The gateway devices are always connected to a power source. The gateways connection acts as a transparent bridge, simply converting RF packets to IP packets and vice versa

4.4.4.1 LoRaWAN

- LoRaWAN is a low power wide area network (LPWAN) specification intended for wireless battery operated things in a regional, national or global network.
- LoRaWAN provides long-range (up to 15km) communication between sensors and base stations, resulting in networks with 2-3x times fewer base stations compared to cellular.

- Fully bidirectional communication enables a wide variety of uses cases requiring uplinks and downlinks: for example, street lighting, smart irrigation, energy optimization or home automation
- LoRaWAN targets key requirements of the internet of things (IoT) such as secure bi-directional communication, mobility and localization services.
- LoRaWAN networks are deployed on cost-free ISM bands allowing any service provider to deploy and operate LoRaWAN networks without having to acquire a license for any frequency.
- The LoRaWAN specification provides seamless interoperability among smart things without the need of complex local installations, enabling the roll out of IoT applications, according to the association.
- In the LoRaWAN network architecture, gateways are connected to the network server via standard IP connections while end-devices use single-hop wireless communication to one or many gateways.
- All end-point communication is generally bi-directional, but also supports operation such as multicast, enabling software upgrades over the air or other mass distribution messages to reduce the on-air communication time.
- LoRaWAN operates in unlicensed radio spectrum. It uses lower radio frequencies with a longer range.
- The LoRaWAN specification defines three device types. All LoRaWAN devices must implement Class A, whereas Class B and Class C are extensions to the specification of Class A devices.
- Class A devices support bi-directional communication between a device and a gateway. Uplink messages (from the device to the server) can be sent at any time (randomly).
- Class B devices extend Class A by adding scheduled receive windows for downlink messages from the server. Using time-synchronized beacons transmitted by the gateway, the devices periodically open receive windows.
- Class C devices extend Class A by keeping the receive windows open unless they are transmitting, as shown in the figure below. This allows for low-latency communication but is many times more energy consuming than Class A devices.
- Fig. 4.4.7 shows LoRaWAN layers.
- The LoRaWAN protocol was defined specifically for LPWAN applications, keeping security, scalable capacity, cost, and ease of deployment in mind.

Applications				
CoAP	MQTT	IPv6/6LoWPAN	Raw	Others
LoRaWAN MAC				
LoRa PHY Modulation				
868 MHz	915 MHz	Other regional bands		

Fig. 4.4.7 : LoRaWAN layers

- The gateways support bidirectional communication and can simultaneously process messages from many LoRa-based sensor nodes.
- A typical LoRaWAN network uses a simple architecture where LoRaWAN base stations connect to the internet through a variety of available backhauls and manage the bidirectional data flow between LoRa sensors and the centralized ThingPark network server provided by Activity.
- Fig. 4.4.8 shows architecture of LoRaWAN.

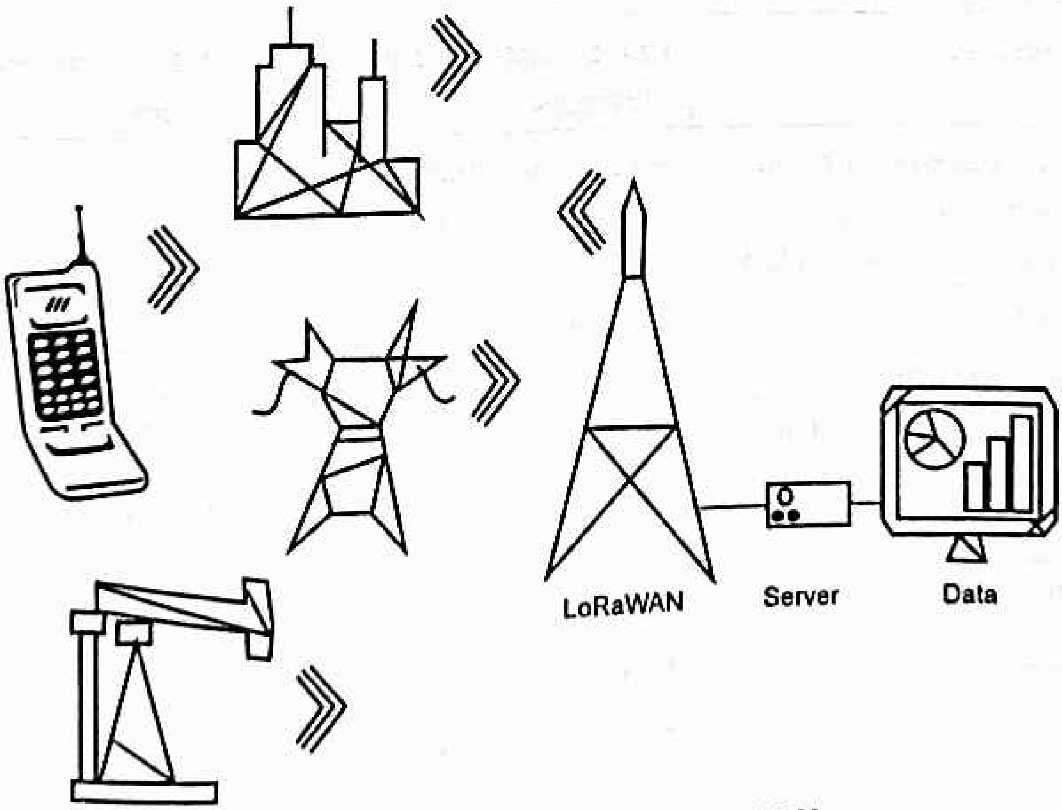

Fig. 4.4.8 : Architecture of LoRaWAN

- End Nodes transmit directly to all gateways within range, using LoRa.
- Gateways relay messages between end devices and a central network server using IP.
- It is designed to allow low-powered devices to communicate with Internet-connected applications over long range wireless connections.
- LoRaWAN messages, either uplink or downlink have a PHY payload composed of a 1-Byte MAC header, a variable byte MAC payload and MIC that is 4 byte in length.
- MAC payload size depends on the frequency band and data rate, ranging from 59 to 230 bytes for the 863-870MHz and 19 to 250 bytes for the 902-928MHz band.
- LoRaWAN endpoints are uniquely addressable through a variety of methods, including the following :
  1. An endpoints can have a global end device ID
  2. An endpoints can have a global application ID
  3. In a LoRaWAN network, endpoints are also known by their end device address, known as a DevAddr a 32-bit address.

❑ Node Type :

Class Type A	Class Type B	Class Type C
Battery Powered	Low Latency	Low Latency
Unicast Messages	Unicast and Multicast Messages	Unicast and Multicast Messages
Server communicates with end device (downlink) during predetermined response windows	Server can initiate transmission at fixed intervals	End-device is constantly receiving
End Device Initiates Communication (Uplink	Extra Receive Window	Server can initiate transmission at any time
Bidirectional with 1 UL+2DL Slot	Bidirectional with Scheduled Downlink Slots	Bidirectional with Most of the Time Listening Mode
• Small Payloads • Long Intervals	• Small Payloads • Long Intervals • Periodic Beacon from Gateway	• Small Payloads

❑ Advantages of LoRaWAN

1. Low Powered sensors that can cover a wide area measured in miles.
2. Operates in the industrial, scientific, and medical (ISM) radio bands. These are free (unlicensed) frequencies, having no upfront licensing cost to use the technology.
3. Low power means long battery life for devices. Sensor batteries can last for two to five years (Class A and Class B).
4. Single LoRa Gateway device is designed to take care of thousands of end devices or nodes.
5. Perfect for monitoring field deployed assets.
6. It is widely used for M2M/IoT applications.
7. LoRaWAN is governed by an alliance.
8. Long-range functionality enables smart solutions, such as smart city applications.
9. Wireless, easy to set up and fast deployment.
10. Security : a layer of security for the network and one for the application with AES encryption.
11. Fully bi-directional communication.

❑ Disadvantages of LoRaWAN

1. Not for large data payloads
2. Has no support for audio or video
3. Limited to Line of Sight (LOS) communication
4. Not for continuous monitoring (except Class C devices).
5. Not ideal candidate for real time applications requiring lower latency and bounded jitter requirements.